

CHARTGENEVAL: TESTING RHYTHM-GAME CHART MEASUREMENTS WITH CONTROLLED ERRORS

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ABSTRACT

Evaluating rhythm-game chart generators is challenging because many different charts can fit the same song, making reference-based comparison inadequate and motivating the use of reference-free metrics. However, such metrics are useful only if we understand which chart properties they actually detect and which irrelevant changes they ignore. We introduce ChartGenEval, a validation framework that evaluates metrics through controlled edits to human-authored charts together with matched controls. Using this framework on Taiko charts, we identify seven reliable metrics spanning timing, pattern, repetition, and density, while showing that several commonly used alternatives can overlook meaningful errors or respond in counterintuitive ways. We further demonstrate how validated metrics can reveal systematic issues in generated charts that conventional evaluation may obscure. Our results suggest that chart-generation metrics should be organized by the properties they measure and reported together with evidence of their sensitivity and invariance.

Index Terms— Music information retrieval, rhythm games, generation evaluation, timing

1. INTRODUCTION

A rhythm-game chart specifies timed player actions, such as center and rim hits in Taiko. Chart generators map music audio to timed symbolic events [1–4], much like drum transcription. Their evaluations often match events to a reference chart using onset-detection precision, recall, and F1 [5]. Unlike a transcription, however, a chart is a design: a generated chart may legitimately add, omit, or rearrange notes for the same song and difficulty. Matching penalizes such additions and omissions even when both designs are suitable. It measures reconstruction, so generation evaluation also needs measurements that reveal specific timing, pattern, and density errors.

Reference-free statistics avoid a target chart. Music-generation studies use distribution, diversity, and repetition statistics [6–8], chart models report perplexity [1], and human studies assess the resulting experience [9]. A plausible statistic can still miss or even reward an error. A shared timing shift preserves note intervals and types (Fig. 2), so chart-only statistics can miss a consistently late chart. Minimizing perplexity favors predictability, while maximizing self-similarity favors repetition; both can reward repetitive edits. Testing metrics with altered examples has precedents: Fréchet Audio Distance is validated with artificial audio distortions [10], Fréchet Music Distance with pitch and velocity changes [11], and SCHmUBERT presents a poor musical example with self-similarity statistics close to its source [12]. STRUM corrects a global offset before scoring onset agreement [13].

Our key observation is that a chart measurement is interpretable only through the errors it has been shown to detect and the changes

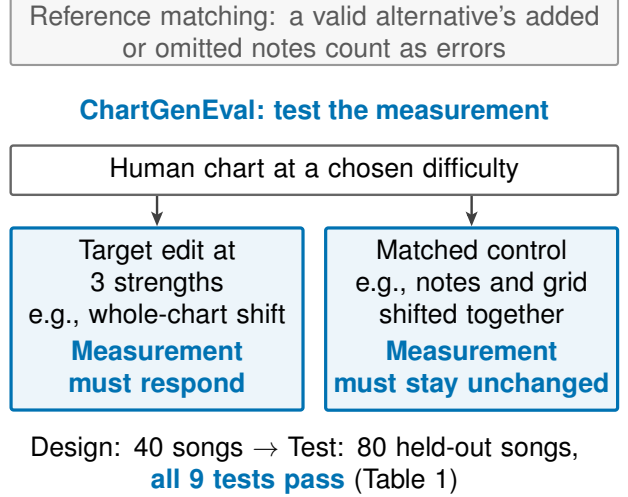


Fig. 1. ChartGenEval tests the measurement, not the chart. Reference matching counts a valid alternative’s added or omitted notes as errors. A measurement passes when it responds to target edits of human charts and stays unchanged under matched controls, such as a center/rim swap or notes and grid shifted together. Controls test invariances, not acceptance of valid alternatives. Tests are fixed on 40 development songs and evaluated on 80 held-out songs (Table 1).

it has been shown to ignore. Motivated by this, ChartGenEval tests each measurement with controlled errors (Fig. 1). Human charts receive edits of increasing strength, such as timing jitter, loop replacement, or burst insertion. A measurement passes when it responds to its target edit and stays unchanged under matched controls. We apply the altered-example approach to chart properties with fixed controls and held-out song tests. Timing is measured against the authored beat grid in the song’s reference chart file, which fixes musical time independently of any chart’s notes; unlike in STRUM, a chart’s shared offset is a timing error. We contribute (i) nine tests linking chart measurements to target edits and matched controls, fixed on development songs before held-out evaluation; (ii) seven measurements of timing, pattern, repetition, and density that pass all nine prespecified tests on 80 held-out Taiko songs (Table 1); and (iii) evidence that some common statistics fail the same tests: post hoc on held-out songs, chart-only statistics miss whole-chart shifts, and repetitive edits lower perplexity and raise self-similarity while the repetition score falls; on development songs, chart offset reveals an early shift in a public generator’s charts. These results support reporting the four properties separately.

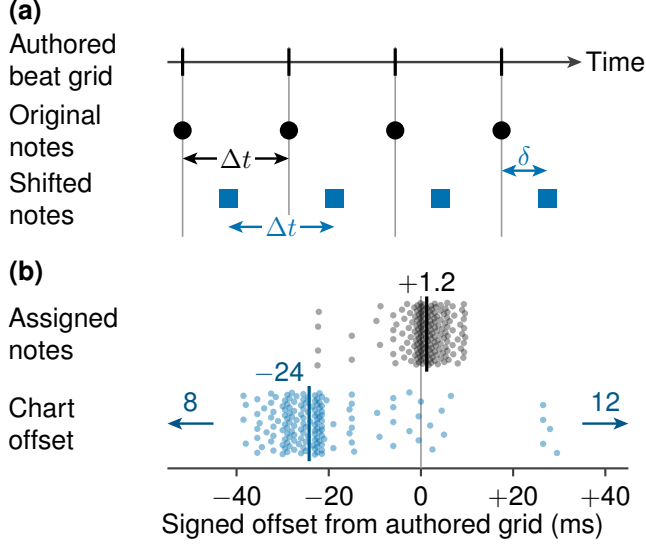


Fig. 2. A shared shift keeps note intervals but moves notes off the authored beat grid. (a) Shifting all notes by δ leaves each interval Δt unchanged; on development songs, such shifts barely change the primary chart-only measurements, whereas chart offset recovers them (Sec. 3.3). (b) Development songs, descriptive: in a public generator’s 170 charts (Sec. 3.4), assigned notes sit near the grid (per-chart median signed offset; median over charts $+1.2$ ms), whereas chart offset shows a shared shift (median -24 ms; human 0 ms). Dots: charts; bars: medians; arrows: charts beyond the axis.

2. METHODS

2.1. Test protocol and data

Each test pairs a measurement with a controlled error. Human charts receive the target edit at three increasing strengths, and matched controls apply changes that should leave the measurement unchanged (Sec. 2.5). For each chart, the first statistic is Spearman correlation between output and error strength (intact plus three strengths); the second is the strongest-edit output minus its matched control. We reverse the signs of timing errors so expected responses to target edits are negative for every measurement. A test passes when both upper bounds of the 99.5% song-bootstrap intervals lie below zero, every control passes, and at least 72 of 80 held-out songs are evaluable. We fixed data, measurements, edits, controls, calibration, thresholds, and analysis of these nine tests before inspecting held-out results.

The corpus contains community-transcribed Taiko charts and associated audio. Training uses 3,880 charts from 813 song groups to fit pattern models and human comparison ranges. The held-out protocol’s language model uses trigrams and add-0.05 smoothing on the full training split. The separate test split contains 120 song groups: the first 40 (170 charts) were used to design measurements and edits; the remaining 80 (333 charts) form the held-out set. Song groups merge aliases using identifiers and audio hashes, rejecting inconsistencies. “Song” below refers to this grouping.

2.2. Timing measurements

Fig. 2 shows why timing requires information outside the generated chart. The grid uses authored bar timestamps, meter, and tempo from

the song’s reference file. This external grid fixes the time origin independently of generated notes. Reference note times and the generator’s grid are unused. For an offset δ ,

$$(t_{i+1} + \delta) - (t_i + \delta) = t_{i+1} - t_i, \quad (1)$$

so intervals alone cannot determine song alignment.

We separate note assignment from timing accuracy. Notes are matched one-to-one to grid points by minimum-cost assignment. Assignment gates depend on grid spacing and lie between 10 and 50 ms. These gates decide which grid point to compare; a tighter tolerance then scores timing. Onset-recovery evaluations similarly allow association windows, for example 20 ms in DDC and 50 ms in GenéLive! [1, 3].

For a matched note time t_i and grid time g_i , define signed offset $e_i = t_i - g_i$ and error beyond tolerance

$$a_i = [|e_i| - 6 \text{ ms}]_+, \quad [z]_+ = \max(z, 0). \quad (2)$$

The 6 ms tolerance is motivated by discrimination of a displaced tone in short, evenly spaced sequences [14]. It is an operating choice whose perceptual validity in gameplay remains to be tested.

On-grid fraction is the fraction of all notes assigned within 6 ms. Unmatched notes stay in its denominator. **Timing tail error** is the 99th percentile of a_i over a chart’s matched notes. It measures excess beyond tolerance after assignment, not raw displacement. We also retain raw offsets and the unmatched-note fraction.

For adjacent matched notes with increasing grid times, interval distortion is $|e_{i+1} - e_i|$. The complementary relative-error summary subtracts a tolerance of 6 ms or 2.5% of the grid interval, whichever is larger, and clips at zero, following the same study’s thresholds [14]. A shared offset cancels; opposite offsets add. For example, $+4$ and -4 ms around a 200 ms interval produce 8 ms distortion and 2 ms excess, although both notes lie within the absolute tolerance. The held-out tail test concerns a_i in Eq. 2.

2.3. Chart-offset estimation

On a dense grid, a shifted note can match a different nearby grid point and still receive a small local error. To estimate the shared shift, we compare candidate offsets against fixed bar, beat, and eighth-note grids G_k . Let $d(x, G_k) = \min_{g \in G_k} |x - g|$. For n note times,

$$S(\delta) = \sum_k \frac{w_k}{n} \sum_i \left[1 - \frac{d(t_i - \delta, G_k)}{12 \text{ ms}} \right]_+, \quad (3)$$

$$\hat{\delta} = \arg \max_{\delta \in [-250, 250] \text{ ms}} S(\delta).$$

Weights are 0.2, 0.3, and 0.5; the search step is 0.5 ms. **Chart offset** is $\hat{\delta}$, with $|\hat{\delta}|$ used in held-out tests. Multiple grid levels reduce periodic ties, though repeated rhythms and syncopation can leave ambiguous offsets. We test recovery for 15, 30, and 60 ms shifts. Tempo alone cannot fix grid phase.

2.4. Pattern and density measurements

We compare each chart with human charts at the same difficulty category (called a *course* in Taiko: Easy, Normal, Hard, Oni, or Ura). A token encodes a note’s type and quantized interval from the prior note. **Pattern familiarity** is $\exp(-r/0.08)$, where r is the fraction of three-token sequences unseen in training at that difficulty.

Table 1. Held-out responses to controlled errors on 80 songs. Rank correlation is with edit strength; strongest-edit change is the strongest edit minus its matched control. Timing tail error and chart offset are sign-reversed, with changes in ms, so negative values indicate the expected response, which need not be monotone (Sec. 3.1). Brackets show Bonferroni-adjusted 99.5% song-bootstrap intervals. A test passes if both upper bounds are negative, every control passes, and at least 72 songs are evaluable; all nine pass, and all 32 controls show zero difference.

Property	Controlled error	Measurement	Rank correlation	Strongest-edit change	Pass
Timing	Timing jitter	On-grid fraction	−0.744 [−0.815, −0.667]	−0.502 [−0.558, −0.446]	✓
	Sparse timing errors	Timing tail error (ms)	−0.234 [−0.344, −0.134]	−0.306 [−0.508, −0.144]	✓
	Whole-chart shift	Chart offset (ms)	−0.916 [−0.981, −0.826]	−55.58 [−61.43, −49.83]	✓
Pattern	Note-type shuffle	Pattern familiarity	−0.800 [−0.856, −0.737]	−0.231 [−0.262, −0.199]	✓
Repetition	Loop replacement	Repetition score	−0.445 [−0.573, −0.312]	−0.165 [−0.226, −0.108]	✓
	Common-pattern rewrite	Repetition score	−0.358 [−0.459, −0.258]	−0.131 [−0.189, −0.078]	✓
	Window shuffle	Repetition score	−0.589 [−0.728, −0.435]	−0.224 [−0.293, −0.156]	✓
Density	Note-rate scaling	Note-rate score	−0.693 [−0.784, −0.579]	−0.582 [−0.672, −0.483]	✓
	Burst insertion	Density-spike score	−0.940 [−0.950, −0.930]	−0.619 [−0.644, −0.593]	✓

For repetition and density, let l and u be the 10th and 90th training percentiles and $h = \max((u - l)/2, 10^{-9})$. Define

$$d(x) = \frac{\max(l - x, 0, x - u)}{h}, \quad s(x) = \frac{1}{1 + d(x)^2}. \quad (4)$$

The score equals one inside the middle 80% of human values and decreases outside it. **Repetition score** applies s to $(M - U)/M$, where M is the number of four-token windows and U is the number of unique windows. **Note-rate score** applies s to note count divided by the time between first and last notes.

Density-spike score measures large local changes separately. It is $\exp(-b/1.5)$, where b is the excess of a chart’s 95th-percentile absolute local density change above its difficulty-specific training limit, in notes/s. Local density uses two-second windows at 0.5-second steps; the limit is the 95th training percentile of that chart-level statistic. Keeping the seven measurements separate prevents a high pattern score from masking a timing error or a burst of notes.

2.5. Controlled errors and controls

Table 1 pairs each edit with its target measurement. Edits create candidate failure cases with known changes in timing, note choice, or density. Strength measures the amount of editing; perceived severity requires player judgments. The nine edits and their three strengths are: **timing jitter**, perturbing all note times by up to $\pm 10/20/30$ ms; **sparse timing errors**, shifting 0.5/1/2% of notes by ± 60 ms; **whole-chart shift**, adding 15/30/60 ms; **note-type shuffle**, swapping center/rim type with probability 0.2/0.4/0.8; **loop replacement**, replacing the first 30/60/100% of note types with a repeated four-note-type pattern; **common-pattern rewrite**, rewriting 30/60/100% of positions toward more likely note types; **note-rate scaling**, multiplying density by 0.5/1.5/2; **burst insertion**, adding 2/4/8 clusters of 8–16 notes at 12–20 notes/s; and **window shuffle**, selecting fixed-duration windows with probability 0.3/0.6/1 and permuting their order.

Pattern edits preserve note times. Window shuffle preserves types and within-window offsets. Its window duration is $4 \cdot 60/B$ seconds for tempo B in beats/min, starting at the first note. These windows need not coincide with authored musical bars; the test addresses reordered windows and local transitions, not musical-form validity. Edits may affect several measurements; the tests establish sensitivity, not exclusive detection.

Controls test identity re-evaluation, a joint one-second shift of notes and the external time origin (grid and duration), and a cen-

ter/rim type swap for timing and density measurements. A 60 ms whole-chart shift is the matched control for chart-only pattern and repetition measurements and an added control for density measurements, whose matched control is the type swap.

2.6. Statistical analysis

Every nonzero edit and control uses five fixed seeds. Replicates are averaged within chart, then complete difficulty variants are averaged within song. Songs are the statistical units, and song-level averages give the reported estimates. Timestamp arithmetic uses integer microseconds. Each measurement has a fixed numerical equivalence tolerance. Constant responses receive zero correlation. For density scaling, strengths are $[0, 0.5, 0.5, 1]$ because halving and increasing density by half have equal departure from the original.

We draw 10,000 bootstrap samples of songs and use Bonferroni-adjusted 99.5% intervals for ten prespecified rows. The original ten rows include algebraically equivalent repetition and variety scores for common-pattern rewriting. Both were required to pass; Table 1 displays their single effective measurement, leaving nine distinct tests of seven measurements. A chart enters only if all required strengths, seeds, and controls are available. Deterministic no-op and missing cells are excluded.

3. RESULTS

3.1. Held-out evaluation

All nine tests in Table 1 meet the prespecified criteria, and all 32 controls pass with an observed difference of zero. Each test includes all 80 songs. Sparse timing errors retain 332 complete charts, window shuffle retains 330, and the other tests retain 333; nineteen no-op cells were excluded.

Two responses require careful reading. The sparse-error change is 0.306 ms on the original larger-is-worse scale because the timing tail error measures excess beyond 6 ms after grid assignment, not the injected 60 ms displacement. A negative rank correlation also permits a nonmonotonic response: mean on-grid fractions under jitter are 0.996, 0.616, 0.408, and 0.495. Re-matching to other grid points may explain this rebound, whose cause remains unresolved; the on-grid fraction detects jitter but is not a monotonic timing loss.

Human-range calibration depends on the underlying feature. In a predeclared secondary held-out analysis, Eq. 4 applied to

Table 2. Common statistics under the held-out tests on 80 songs (post hoc, not prespecified). Columns 1–9: edits in Table 1 row order. ✓: passes its rule; ○: unchanged at every seed and strength; ×: moves toward the conventionally better value (lower perplexity; higher self-similarity, if read as better), both intervals excluding zero; –: unchanged by design (time-only statistic, type-only edit).

Statistic	1	2	3	4	5	6	7	8	9
Perplexity	✓	✓	○	✓	×	×	✓	✓	✓
Self-similarity	✓	✓	○	✓	×	×	✓	✓	✓
IOI-histogram JS	✓	✓	○	–	–	–	✓	✓	✓
Onset F1, 50 ms	○	✓	✓	–	–	–	✓	✓	✓
Onset F1, 20 ms	✓	✓	✓	–	–	–	✓	✓	✓

language-model loss fails to identify the strongest common-pattern rewrite (change +0.021, nominal 95% interval $[-0.034, 0.077]$).

3.2. Common statistics under the same tests (post hoc)

Post hoc, five common statistics took all nine tests (Table 2; seven more were also computed); each ChartGenEval measurement got only its target test. Perplexity fell from 8.92 to 7.07 under loop replacement, which is model-free, and to 5.89 under common-pattern rewrite, selected and scored by the same model; three separately fitted scorers fell 3.26–3.50 (upper bounds ≤ -2.64). Self-similarity rose from 0.103 to 0.154 and 0.144, rewarding both only if higher is read as better; distinct-4 detects both yet rises under note-type shuffle (+0.21 [+0.20, +0.23]). Maximum-matching 50 ms onset F1 [5] stayed 1.000 at ± 30 ms jitter.

3.3. Timing-offset recovery on development songs

Shifting notes by 15, 30, or 60 ms changes no primary chart-only measurement by more than 0.03 human standard deviations, and with the authored grid the chart-offset estimate moves by the injected shift to within 3 ms for 169 of 170 charts.

3.4. Public generator outputs on development songs

Of the public generators we ran, two output Taiko note types. We omit TaikoNation [2], whose notes after our conversion aligned with the authored grid near chance level. Mapperatorinator v32 [15] produced 170 charts for the 40 development songs (courses requested as osu! star ratings 2.0–7.5); these outputs were seen during measurement design, so the readings are descriptive. The median chart offset is -24 ms (Fig. 2b; interquartile range -29 to -19 ms; human 0 ms), for reasons we did not determine, whereas assigned notes have a median signed offset of $+1.2$ ms (median over charts). From Easy to Ura, note rates are a median 2.2, 2.3, 1.9, 1.3, and 1.1 times the paired human chart’s, and median note-rate scores are 0.18, 0.11, 0.19, 1.00, and 1.00 (human 1.00); median pattern familiarity is 0.05, 0.01, 0.54, 0.76, and 0.78 (human 0.66, 0.69, 0.71, 0.81, and 0.75).

4. DISCUSSION

Lower perplexity indicates easier prediction, which common-pattern rewriting achieves by reducing note-choice diversity. Self-similarity describes repeated material [8], so its increase under loop replacement is also consistent with what it measures. These examples expose the ambiguity of treating either direction as improved generation quality [16]. Human-range calibration of language-model loss

also failed to detect the rewrite, so testing the intended interpretation of a measurement is necessary even after calibration.

Local and shared timing measurements answer different questions. The on-grid fraction summarizes small assigned errors, the timing tail error captures rare excesses, and chart offset estimates the shared displacement that grid assignment can hide (Sec. 3.4).

In practice, a generator evaluation can report onset F1 for reconstruction alongside the seven measurements, placing raw offsets and the unmatched-note fraction beside the tolerance-based timing scores.

The tested measurements are candidate rewards and constraints for reinforcement learning toward human-like charts with fewer perceptible errors. Tests with generator outputs should check whether score gains preserve useful note choices, and player studies should establish perceptual benefits.

4.1. Limitations

The held-out study applies fixed edit families to new songs from one Taiko corpus. Measurements and edits were designed together on development songs, so responses to unseen generator errors remain to be tested, and the controls check specified invariances rather than acceptance of every valid alternative chart. Player judgments, wrong-song pairings, and removal of musically important notes lie outside this study. Human comparison ranges pool broad course categories, and song-bootstrap intervals treat songs as exchangeable without modeling shared chart authors or sources.

Timing requires a reliable external grid. Repeated rhythms can produce multiple plausible offset estimates, and the 6 ms tolerance awaits validation in gameplay. Held-out results cover only the fixed tolerances, gates, and grid weights reported here. The on-grid fraction has no reference-note recall term, so deleting off-grid notes can raise it; it should be read with note rate and unmatched-note fraction.

4.2. Data and code availability

Code, evaluation summaries, and plotting scripts are available at <https://github.com/JacobLinCool/ChartGenEval>. These materials support checking the reported summary results. End-to-end regeneration of the frozen held-out run additionally requires its exact source snapshot and raw records, which the release does not include. Source charts and audio are not redistributed.

4.3. Conclusions

ChartGenEval links seven chart measurements to specified errors and controls. Nine tests on 80 held-out Taiko songs establish their target responses to edits of human charts under a fixed protocol; unseen generator errors, timing without authored grids, acceptance of valid alternative charts, and player judgments remain untested. In post hoc held-out tests, perplexity and self-similarity favor repetitive edits and chart-only statistics miss timing shifts; on development songs, a generator’s charts run early. The evidence supports reporting timing, pattern, repetition, and density separately, each interpreted through the errors it has been tested to detect.

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6. COMPLIANCE WITH ETHICAL STANDARDS

This study analyzes community-transcribed charts, associated audio, and derived numerical records. It involved no recruited participants, player profiles, or gameplay records. Potentially copyrighted source charts, audio, and generated charts are not redistributed. High-volume deployment of automated generation requires platform policy and moderation. No external funding, sponsorship, or donated computing resources supported this study. The authors declare no conflicts of interest.

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